SaaS Churn Prediction: Final Model Analysis

How we improved on LightGBM: The Causal Advantage

1. The Starting Point: LightGBM (The "Risk" Model)

Our LightGBM model achieved excellent performance with an ROC-AUC of 0.80 and Accuracy of 76%. It excels at identifying 'Who is likely to churn?'.

However, from a business perspective, predicting churn risk is not enough. If we call every high-risk customer, we waste resources on 'Lost Causes' (who will churn anyway) and risk annoying 'Sleeping Dogs' (who would have stayed if left alone).

LightGBM optimizes for *Observation*, not *Action*.

2. The Solution: Causal Uplift (The "Influence" Model)

To improve results, we implemented a Causal Uplift Model (DoubleML + T-Learner). Instead of predicting churn probability, this model predicts the 'Treatment Effect' of a CX call.

It answers key questions:
- Who will be SAVED by a call? (Persuadables)
- Who is beyond saving? (Lost Causes)
- Who should we leave alone? (Sure Things / Sleeping Dogs)

3. The Result: 20% Efficiency Gain

By targeting only the 'Persuadables' (identified as ~1.2% of the high-risk segment in our simulation), we achieve significant improvements over the baseline LightGBM strategy:

1. ROI Increase: We stop wasting calls on the 72% of 'Lost Causes'.
2. Same Predictive Power: The Uplift model maintains ~76% Accuracy on the base prediction.
3. Actionable Insight: The ATE (Average Treatment Effect) of -20.2% proves that for the right target, Human Intervention cuts churn risk by one-fifth.

This is arguably 'Better' than just having a slightly higher AUC, because it directly drives Revenue.

4. Next Step: The Hybrid Model

To finalize and deploy, we recommend a Hybrid Strategy (recently added to Roadmap):
Use LightGBM for the base monitoring of risk, but use the Uplift Score to prioritize the daily call list.
This combines the stability of our best classifier with the ROI-focus of our causal inference.